

Maths — UC02 Percentages: Key Notes

數學 UC02 百分比 重點筆記 · Test: Tuesday 13 October 2026 (Week A), 40 minutes, calculators
NOT allowed

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 **Source note 資料來源:** These notes are built from Steve's **marked Official Homework 2** — the teacher's real question types and methods (red-pen marked 9/9). Ms Leung posts **no lesson slides or worksheets online** — all teaching uses the physical **Booklet 02**, so there are no teacher slides to check against. Questions below mirror the homework's style exactly.

本筆記根據已批改的 Official Homework 2 (實題題型) 整理；老師沒有在網上發佈課件或工作紙（用實體 Booklet 02 上課）。

1. The Three-Way Conversion 分數、小數、百分比互換

Fraction $\rightleftharpoons$ Decimal $\rightleftharpoons$ Percentage. Write conversions as a **chain**, like the teacher does:

5/8 $\rightarrow$ 0.625 $\rightarrow$ 62.5% | 1 1/5 $\rightarrow$ 1.2 $\rightarrow$ 120%

像老師一樣把換算寫成一串鏈。

Fraction 分數	Decimal 小數	Percentage 百分比	How 做法
5/8	0.625	62.5%	$5 \div 8 = 0.625$; $\times 100 = 62.5\%$
76/100 $\rightarrow$ 19/25	0.76	76%	Simplify FIRST 先約簡 , then convert
13/20	0.65	65%	$13 \div 20 = 0.65$
1 1/5 (mixed 帶分數)	1.2	120%	Percentages can be over 100% !
42/100 $\rightarrow$ 21/50	0.42	42%	Simplify FIRST 先約簡

★ **Rule 1 — Simplify first 先約簡** : if the fraction can be reduced (76/100 $\rightarrow$ 19/25), do that before converting.

★ **Rule 2 — Chain rule 鏈式** : fraction $\rightarrow$ decimal $\rightarrow$ percentage, all on one line.

2. Ordering — Convert to Decimals First 比大小：先化為小數

To order a mixed set **smallest $\rightarrow$ largest**, convert everything to **decimals first**, then compare. Watch out for **recurring decimals 循環小數** (a dot over the repeating digit).

把所有數先化成小數再比較；注意循環小數。

- Example A: **5/7, 0.64, 71%, 0.71 $\dot{}$**
 $\rightarrow$ **0.714285..., 0.64, 0.71, 0.717171...**
 $\rightarrow$ order: **0.64 < 71% < 5/7 < 0.71 $\dot{}$**
(注意：0.71 $\dot{}$ = 0.71717..., 比 5/7 = 0.71428... 稍大！)

- Example B: **1/1000, 11/1000, 0.11%, 0.0108**
 $\rightarrow$ **0.001, 0.011, 0.0011, 0.0108**

→ order: $1/1000 < 0.11\% < 0.0108 < 11/1000$

(小心: $0.11\% = 0.0011$, 比 0.0108 還小!)

★ **Trap 警告**: 0.11% is NOT 0.11 — it is $0.11 \div 100 = \mathbf{0.0011}$. And $0.71\bar{1}$ ($0.7171\dots$) $> 5/7$ ($0.7142\dots$). Write out 4–5 decimal places before deciding!

3. Finding a % of a Quantity 求某數量的百分之幾

Turn the percentage into a **decimal multiplier**, then multiply:

$$44 \times 17.5\% = 44 \times 17.5/100 = 44 \times 0.175 = 7.7$$

把百分比變成小數乘數，再乘。

- $62.5\% \text{ of } 96 = 96 \times 0.625 = 60$
- $130\% \text{ of } 45 = 45 \times 1.3 = 58.5$ (超過 100% 也照乘)

4. Percentage Increase & Decrease 百分比增減

Increase 加: multiply by $(100\% + \text{rate})$. **Decrease 減**: multiply by $(100\% - \text{rate})$.

$$\text{chair } \$200, +15\% \rightarrow \$200 \times (100\% + 15\%) = \$200 \times 115\% = \$200 \times 1.15 = \$230$$

加: 乘以 $(100\% + \text{比率})$; 減: 乘以 $(100\% - \text{比率})$ 。

- Increase: 16% more than 500 seats $\rightarrow 500 \times 116\% = 500 \times 1.16 = 580$
- Decrease by subtracting: 20% of 25 questions wrong $\rightarrow 25 \times 0.20 = 5$ wrong, so $25 - 5 = 20$ correct
也可以先算出減少了多少，再減去。
- Decrease by multiplier: bike \$880, $-15\% \rightarrow 880 \times (100\% - 15\%) = 880 \times 85\% = 880 \times 0.85 = 748$

★ The teacher accepts **both** decrease methods: $(100\% - \text{rate})$ multiplier OR find the amount first and subtract. Use whichever you find easier — but show working.

5. "X as a % of Y" X 佔 Y 的百分之幾

$$(X \div Y) \times 100\%$$

$$15,300 \text{ story books out of } 17,000 \rightarrow 15,300 \div 17,000 \times 100\% = 0.9 \times 100\% = 90\%$$

6. Two-Step Money Problems 兩步金錢應用題

Work step by step, **one line per step**, and finish with a **∴ conclusion sentence** — the teacher expects this!

Example: \$250, spends 30% on books, then pays \$145 for a school bag.

Step 1: $250 \times 30\% = 250 \times 0.3 = 75$ (spent on books)

Step 2: $250 - 75 = 175$ (left after books)

Step 3: $175 - 145 = 30$ (left after bag)

∴ She has \$30 left.

一步一行，最後一定要寫 ∴ 結論句，例如 "∴ The new price of the chair is \$230."

★ **Test checklist 考試檢查表：** ① Simplify fractions first 先約簡 ② Write conversion chains 鏈式寫法 ③ Convert mixed sets to decimals before ordering ④ Use (100% ± rate) multipliers for increase/decrease ⑤ One line per step in word problems ⑥ **Always end with a ∴ conclusion sentence** ⑦ No calculator — practise decimal × decimal by hand!